

High-pressure processing when heat is not the kill step

A set-point is not a pasteurisation number — log, law, and quality are separate readings of the same cycle.

1. Condition
Packed food under isostatic 300–600 MPa, seconds to minutes, usually chilled or ambient. Piece size does not create a cold centre.



2. Mechanism (graded)
Reviews compile membrane leak, lipid-phase change, and non-covalent protein unfolding. Ice and lactate change the mechanical or solute coefficients. Not one proven molecular cause for every log.



3. What is measured
Colonies, enzyme activity, or texture. Selective agar and immediate sampling can miss injury. No colony is not automatically death.



4. Three readings
Hurdle: refrigerated vegetative kill at 500–600 MPa in high-aw foods.
Law: industrial milk HPP is not legal heat pasteurisation on the included comparison; juice APC is not a named-pathogen 5-log.
Spores: ambient commercial HPP is not a botulinum cook — HPTP or a cold chain.

Quality (polyphenol oxidase, bioactive proteins, kelp hardness) can move against plate counts on the same cycle. Ultrasound, plasma, and irradiation share a label with pressure, not a mechanism. Magnitudes stay in Tables 1–3 of the article; this panel does not pool them.